

HAMI, China

WMO: 522030

Lat: 42.817N

Lon: 93.517E

Elev: 2425

StdP: 13.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) 0.1	(d) -14.3	(e) 2.8	(f) 11.3	(g) -10.7	(h) 3.4	(i) 12.1	(j) 8.4	(k) 20.5	(l) 7.5	(m) 19.1	(n) 2.5	(o) 40	(p) 0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.5	(c) 99.3	(d) 66.8	(e) 96.7	(f) 65.9	(g) 94.3	(h) 65.0	(i) 70.9	(j) 91.8	(k) 69.1	(l) 90.1	(m) 67.5	(n) 88.1	(o) 3.3	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	99.2	79.7	62.1	91.3	79.0	60.1	84.9	78.7	36.6	91.6	34.9	90.1	33.5	88.0	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.2	(b) 10.3	(c) 8.1	(e) -6.7	(f) 104.1	(g) 5.2	(h) 2.0	(i) -10.5	(j) 105.5	(k) -13.5	(l) 106.7	(m) -16.5	(n) 107.8	(o) -20.2	(p) 109.3		
(a) 13.2	(b) 10.3	(c) 8.1	(e) -7.5	(f) 74.4	(g) 5.0	(h) 2.8	(i) -11.1	(j) 76.4	(k) -14.0	(l) 78.1	(m) -16.8	(n) 79.6	(o) -20.5	(p) 81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	15.0	27.8	43.5	58.8	69.2	78.2	81.3	78.4	67.1	52.2	35.6	19.1		
	DBStd	23.77	6.28	7.35	8.76	8.37	6.86	5.42	4.84	5.11	6.29	7.30	7.89	7.15		
	HDD50	3429	1086	622	239	24	0	0	0	0	1	64	434	959		
	HDD65	6270	1551	1042	666	216	39	1	0	1	48	400	882	1424		
	CDD50	4262	0	0	38	287	595	845	969	881	514	132	1	0		
	CDD65	1628	0	0	0	30	168	395	504	417	112	2	0	0		
	CDH74	20109	0	0	3	418	2046	4819	6365	4959	1407	92	0	0		
	CDH80	10134	0	0	0	93	804	2487	3581	2641	517	11	0	0		
	WSAvg	3.2	2.6	3.1	4.1	4.4	4.0	3.4	3.1	2.9	2.7	2.5	2.6	2.5		
	PrecAvg	1.5	0.0	0.0	0.0	0.1	0.1	0.3	0.3	0.2	0.1	0.1	0.1	0.0		
(15) Precipitation	PrecMax	3.1	0.4	0.4	0.6	0.8	0.6	1.3	1.2	0.9	0.5	0.7	0.6	0.6		
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.7	0.1	0.1	0.1	0.2	0.2	0.4	0.3	0.2	0.1	0.2	0.1	0.1		
	PrecStd	0.7	0.1	0.1	0.1	0.2	0.2	0.4	0.3	0.2	0.1	0.2	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	52.7	73.7	87.2	94.9	99.9	103.5	101.6	94.1	81.5	62.8	41.2		
		MCWB	27.8	37.3	48.1	56.1	61.4	65.5	68.7	67.5	62.9	55.2	45.1	32.1		
	2%	DB	31.9	48.3	68.9	82.6	91.2	96.8	100.3	98.0	89.6	76.4	57.4	36.4		
		MCWB	25.0	35.1	45.9	53.6	59.7	64.3	68.1	66.1	61.0	53.0	41.9	28.6		
	5%	DB	28.9	44.6	64.6	78.9	87.9	94.5	97.5	95.3	85.9	72.1	53.2	32.9		
		MCWB	22.8	32.9	43.9	52.1	58.6	64.0	67.1	65.3	59.7	51.0	40.0	26.4		
	10%	DB	26.1	40.9	60.1	75.2	84.6	91.8	94.7	92.4	82.2	67.8	48.6	29.7		
		MCWB	20.9	30.7	41.8	50.6	57.2	63.2	66.3	64.7	58.3	49.0	37.6	24.2		
	0.4%	WB	28.2	37.9	49.1	57.4	64.7	70.9	74.1	72.9	66.3	56.4	45.6	32.5		
		MCDB	35.4	51.1	70.6	83.8	88.7	91.3	93.2	92.7	85.4	77.6	61.6	40.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.3	35.5	46.6	55.0	61.9	68.4	71.6	70.2	63.6	53.9	42.6	29.0		
		MCDB	31.3	47.2	67.1	79.8	87.4	89.3	92.9	90.1	83.5	73.4	56.0	35.7		
	5%	WB	23.1	33.2	44.5	53.0	59.9	66.6	69.9	68.3	61.6	51.9	40.3	26.6		
		MCDB	28.6	43.9	63.6	76.5	84.5	88.0	91.8	88.1	81.6	69.4	52.4	32.5		
	10%	WB	21.0	31.0	42.1	51.2	58.0	65.1	68.3	66.7	59.7	49.9	37.9	24.5		
		MCDB	25.8	40.2	59.5	73.3	81.6	86.8	89.5	86.9	79.5	66.1	48.2	29.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	23.5	25.3	25.1	25.1	24.4	24.5	26.7	29.6	28.5	21.6	19.1		
		MCDBR	22.5	26.0	29.2	29.1	29.8	28.1	28.6	30.7	33.1	32.8	26.5	20.6		
		MCWBR	17.0	16.0	14.8	12.4	12.5	12.0	11.3	12.0	13.0	14.6	14.9	14.6		
		MCDBR	22.0	24.9	27.6	26.6	27.2	24.8	25.2	27.4	30.4	31.5	26.0	20.0		
	5% WB	MCWBR	16.6	15.5	14.2	11.2	12.3	11.9	11.6	12.3	12.9	14.2	14.7	14.3		
		MCWBR	16.6	15.5	14.2	11.2	12.3	11.9	11.6	12.3	12.9	14.2	14.7	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.382	0.428	0.474	0.477	0.488	0.480	0.449	0.426	0.400	0.364	0.334		
	taud		2.105	2.016	1.912	1.861	1.905	1.958	2.018	2.083	2.066	2.046	2.084	2.110		
	Ebn at Noon		243	249	252	252	255	252	252	257	252	239	228	229		
	Edn at Noon		33	43	54	61	60	57	54	49	46	41	34	30		
(44) All-Sky Solar Radiation	RadAvg		690	1023	1510	1897	2213	2223	2171	1991	1690	1238	795	601		
	RadStd		65	53	63	101	100	141	91	78	61	40	46	59		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.83	N/A	N/A	+1.17	-1.84	-2.63	N/A	N/A	+263	+182
(47) Regional (3 neighbors)		+1.19	N/A	N/A	+1.37	N/A	-1.66	N/A	N/A	+263	+182

Nomenclature: See separate page